Cybersecurity Awareness

CSE212 Cybersecurity – Assignment Part 1

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Winter 2026

Introduction to Cybersecurity

- Cybersecurity refers to the practice of protecting computers, networks and digital information from cyber attacks and unauthorized access.

- In the modern digital world, almost every activity such as banking, communication, shopping, and education depends on the internet. This makes digital systems a major target for cyber criminals.

- Cybersecurity uses technologies such as firewalls, encryption, and antivirus software to secure digital systems and prevent cyber attacks..

CIA Triad in Cybersecurity

- The CIA Triad forms the foundation of cybersecurity principles.

- Confidentiality – ensures sensitive information is accessible only to authorized users.
- Integrity – ensures data remains accurate and cannot be modified by unauthorized users.
- Availability – ensures systems and data remain accessible when required.

Cybersecurity Awareness

- Cybersecurity awareness refers to educating individuals about cyber threats and safe online behavior.

- Many cyber attacks occur because users unknowingly click malicious links or download infected files.

- Cybersecurity awareness includes:
  - recognizing phishing emails
  - creating strong passwords
  - protecting personal information online
  - avoiding suspicious downloads

Importance of Cybersecurity Awareness

- Cybersecurity awareness is important because cybercrime is increasing rapidly across the world.

- Organizations and individuals store sensitive information online including financial data, personal records, and business information.

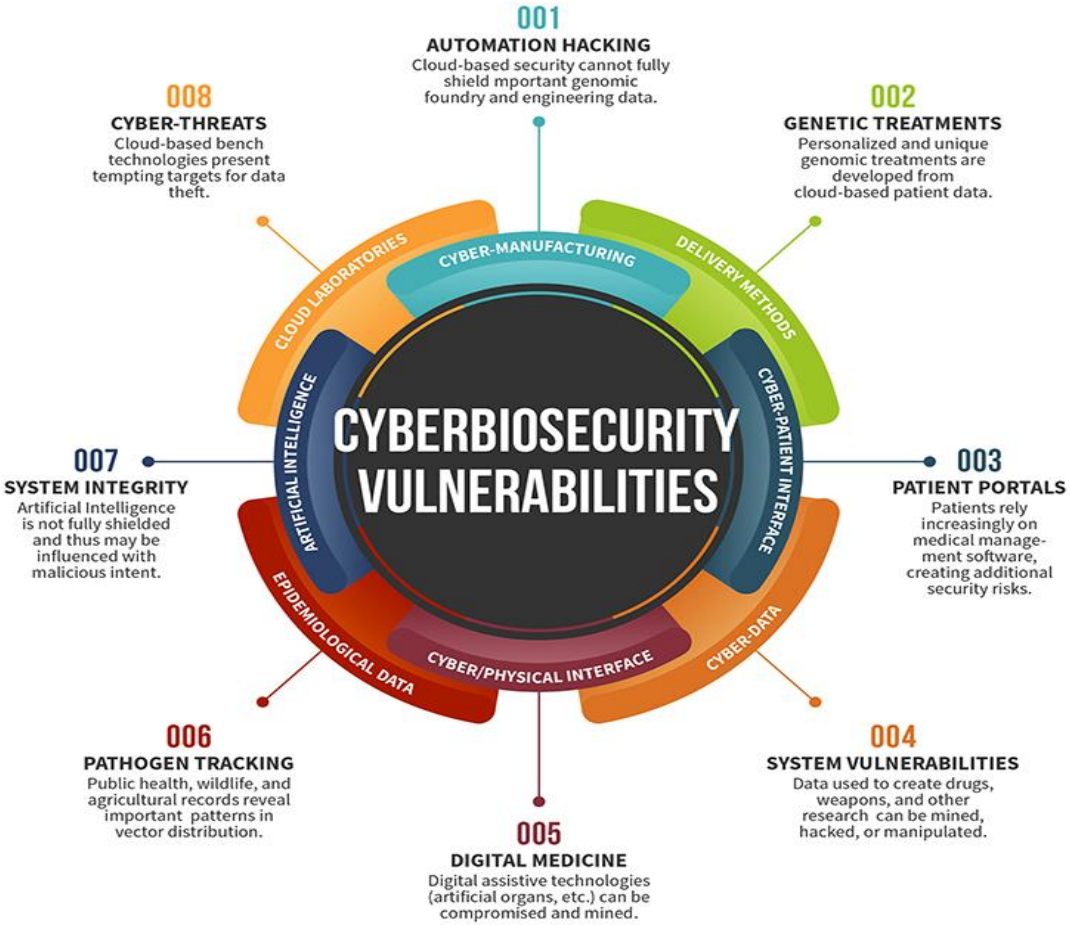
Benefits of cybersecurity awareness include:

- protection of personal data
- prevention of identity theft
- reduction of financial fraud
- improved digital safety

Major Cybersecurity Threats

- Cybersecurity threats are malicious activities designed to damage systems or steal sensitive information.

- Examples include:
  - Malware
  - Phishing
  - Ransomware
  - Social Engineering
  - DDoS attacks
  - Data breaches

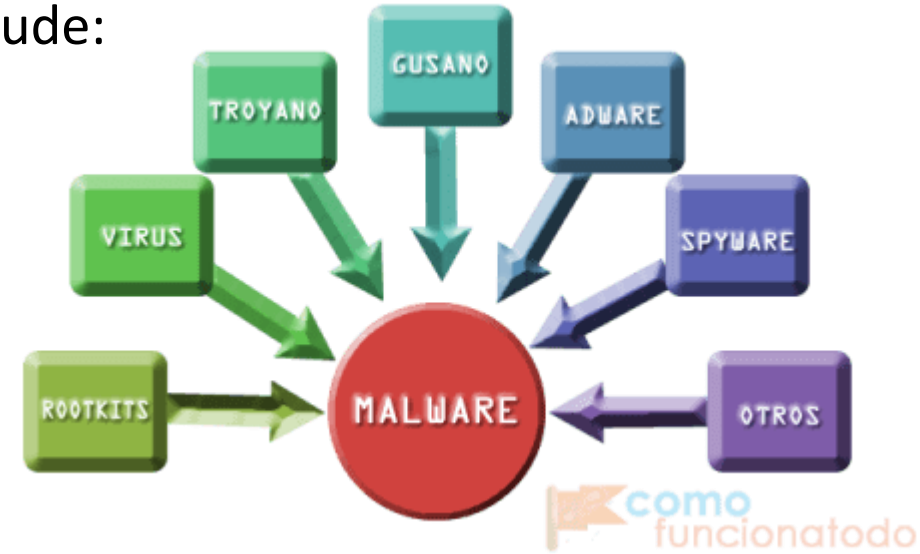


Malware

- Malware is malicious software designed to harm computer systems or steal information.

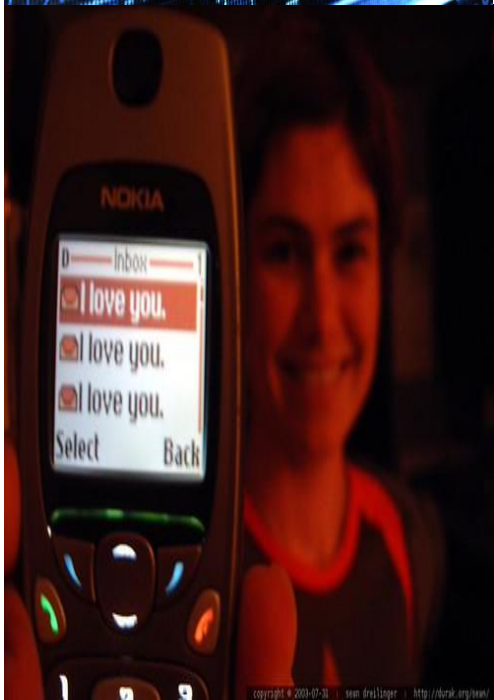
- It spreads through infected email attachments, malicious downloads, compromised Malware is malicious software designed to harm computer systems or steal information.

- It can spread through infected email attachments, malicious downloads, compromised websites, or USB devices.
- Common types of malware include:
- • Virus
  • Worm
  • Trojan Horse
  • Spyware
  • Ransomware
- sed websites, or USB devices.



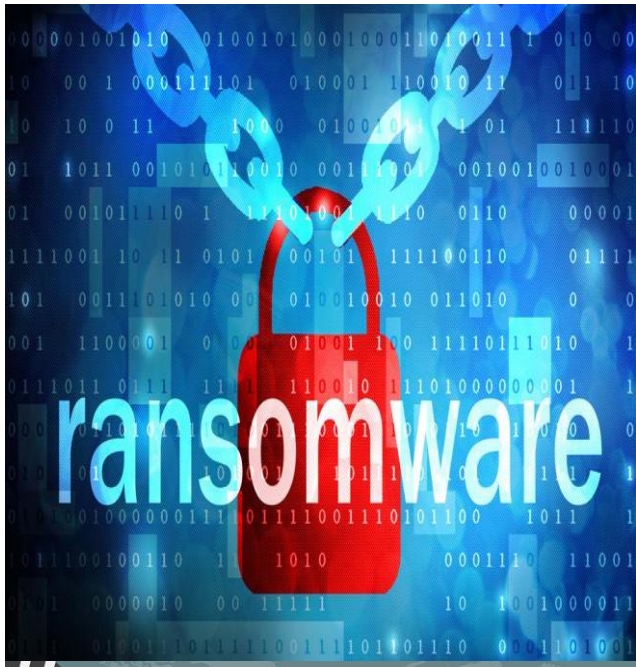
Virus Example – ILOVEYOU Virus

- The **ILOVEYOU virus** discovered in 2000 was one of the most destructive computer viruses.
- It spread through email attachments disguised as a love letter.
- When users opened the attachment, the virus spread automatically to all contacts in the email list.
- Impact of the attack:
- • infected millions of computers worldwide
  • caused nearly **$10 billion in damages**
  • disrupted businesses and government systems



Ransomware

- Ransomware is a type of malware that encrypts files or locks computer systems.
- Attackers demand a ransom payment in exchange for restoring access to the files.
- These attacks often target hospitals, government systems, and businesses because they rely heavily on data availability.
- Payments are usually requested through cryptocurrency to hide the attacker's identity.



WannaCry Ransomware Attack (2017)

- WannaCry was one of the largest ransomware attacks.
- It exploited a Windows vulnerability called EternalBlue.
- Impact:
  - infected more than 200,000 computers
  - spread across 150+ countries
  - affected hospitals and companies

NotPetya Attack (2017)

- NotPetya was a destructive cyber attack disguised as ransomware.

- It spread through infected software updates and disrupted many international companies.

- Major consequences:
  - disrupted global shipping company **Maersk**
  - impacted multiple international businesses
  - caused approximately **$10 billion in damages**
- NotPetya demonstrated how cyber attacks can severely impact global industries.
- It caused approximately $10 billion in global damages.

Phishing

- Phishing is a cyber attack in which attackers impersonate trusted organizations to trick users into revealing sensitive information.

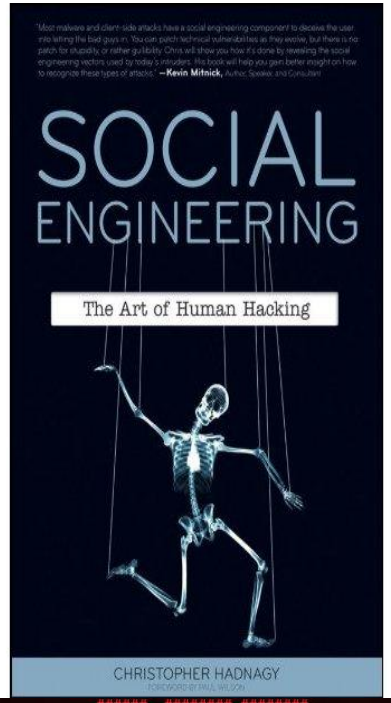
- Phishing attacks commonly occur through:
  - fake emails from banks or companies
  - fake login websites
  - fraudulent SMS messages

- Users who click malicious links may unknowingly share passwords or financial information with attackers.

- Phishing remains one of the most common cyber threats.

Social Engineering (Twitter Hack 2020)

- Social engineering attacks manipulate people into revealing confidential information rather than exploiting technical vulnerabilities.
- In 2020, hackers used social engineering techniques to gain access to internal Twitter tools.
- They took control of several high-profile accounts including Elon Musk, Bill Gates, and Barack Obama.
- The attackers posted cryptocurrency scam messages from these accounts.



DDoS Attack and Data Breach Example

- A Distributed Denial of Service (DDoS) attack floods a server with large amounts of traffic, making websites unavailable.
- Example: Dyn DNS attack (2016) disrupted major services such as Twitter, Netflix, and Reddit using the Mirai botnet.
- Another major cyber attack example is the Equifax data breach (2017).
- Hackers stole personal data of 147 million people, including Social Security numbers and credit information.

Cybersecurity Best Practices & Conclusion

- To improve cybersecurity, individuals should follow important security practices such as:
  - using strong and unique passwords
  - enabling two-factor authentication
  - regularly updating software
  - installing reliable antivirus software
  - avoiding suspicious emails and links

- By understanding cyber threats and following safe online practices, individuals and organizations can significantly reduce the risk of cyber attacks.

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